Radian measure

1

a State the multiplier to convert an angle from degrees to radians.

b State the multiplier to convert an angle from radians to degrees.

2 Convert the following to radians, correct to two decimal places where applicable:

a 360° **b** 180° **c** 90° **d** 225°

e -300° **f** 112° **g** 29° **h** 161.17°

i $321^\circ 31'$ **j** -45° **k** 45° **l** 330°

m -60° **n** 120°

3 Determine whether the following statements are true or false. If it is false, correct the statement.

a If $180^\circ = \pi$ radians, then 90° must be equal to $\frac{\pi}{2}$ radians.

b 60° must be equal to $\frac{\pi}{6}$ radians, because it is $\frac{1}{6}$ of 360°.

c 30° must be equal to $\frac{\pi}{3}$ radians, because it is half of 60°.

d 210° must be equal to $\frac{7\pi}{6}$ radians, because it is seven groups of 30°.

e 45° must be equal to $\frac{\pi}{4}$ radians, because it is half of 90°.

f 225° must be equal to $\frac{7\pi}{4}$ radians, because it is seven groups of 45°.

4 Convert the following to degrees:

a $\frac{\pi}{3}$ **b** $\frac{\pi}{6}$ **c** $\frac{\pi}{4}$ **d** $\frac{3\pi}{2}$

e $\frac{5\pi}{4}$ **f** $-\frac{2\pi}{3}$ **g** $-\frac{5\pi}{3}$ **h** 4.2 radians

5 For each of the following angles:

i Find the complement. **ii** Find the supplement.

a 30° **b** $\frac{\pi}{4}$ radians **c** $89^\circ 30'$ **d** $\frac{\pi}{11}$ radians

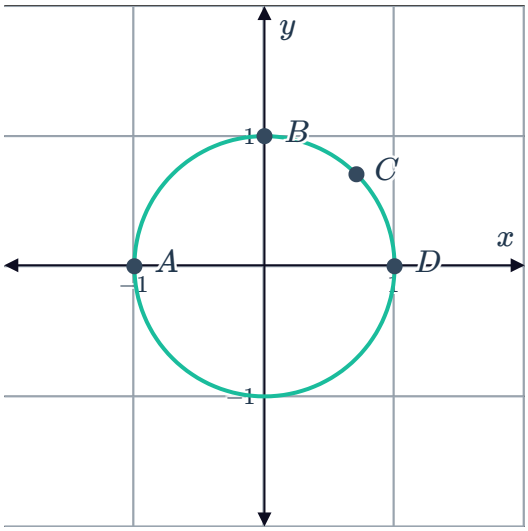
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Unit circle

6 Determine the quadrant of the unit circle in which the following angles lie:

a 2.6 radians **b** $-\frac{5\pi}{6}$ radians **c** $\frac{\pi}{6}$ radians **d** -6.4 radians

7 The graph shows a number of points on the unit circle:
Name the point that corresponds to the following angle around the circle:



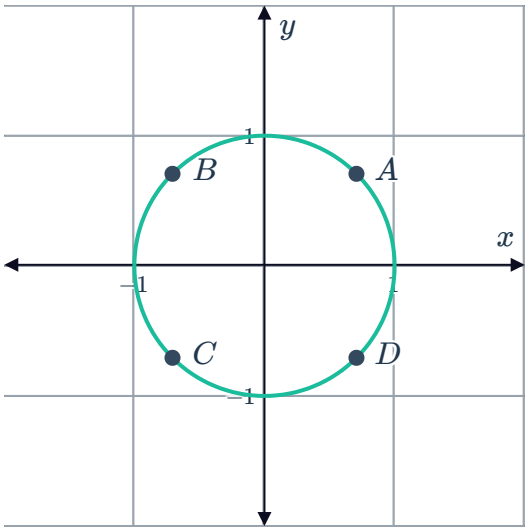
a $\frac{\pi}{2}$ units

b π units

c -2π units

d $-\frac{7\pi}{4}$ units

8 The graph shows a number of points on the unit circle:
Name the point that corresponds to the following angle around the circle:



a $\frac{\pi}{4}$ units

b $\frac{3\pi}{4}$ units

c $\frac{21\pi}{4}$ units

d $\frac{-9\pi}{4}$ units

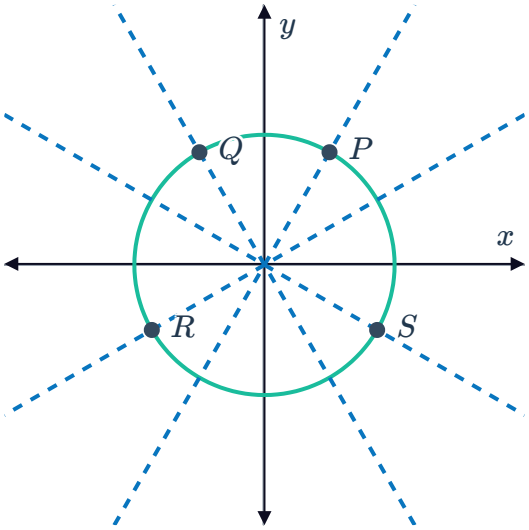
9 Given that π^c represents half a circle, state the fraction of the the unit circle represented by the following angles:

a $\frac{\pi^c}{2}$ **b** $\frac{2\pi^c}{3}$ **c** $\frac{4\pi^c}{7}$ **d** $5\pi^c$

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10 The graphs shows the unit circle divided into 12 equal segments, and a number of points on the circumference:
State two angles between -2π and 2π that correspond to the following points:

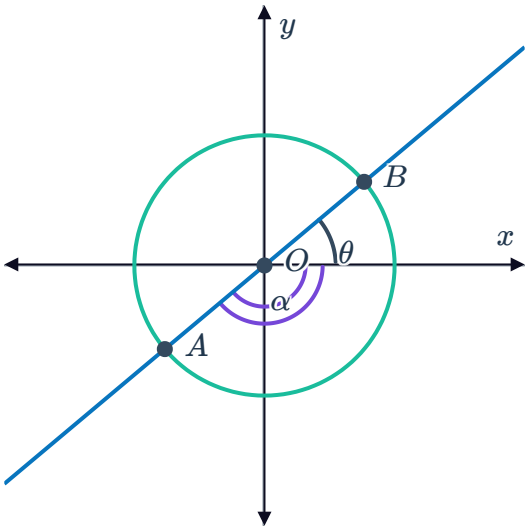
a R **b** P **c** S **d** Q



11 The graph shows a line through two points on the unit circle, A and B, and the origin, O. The line segment OB forms an angle of $\theta = \frac{\pi}{13}$ with the positive x-axis. The angle between the line segment OA and the positive x-axis is α.
Find the value of α when:

a $0 \leq \alpha < 2\pi$

b $-2\pi < \alpha \leq 0$



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